NAIM BRO

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RESEARCH INTERESTS

Political elites and social stratification; surnames, kinship and long-run social mobility; legislative behaviour and political networks; large language models as instruments of social measurement; computational social science.

CURRENT POSITIONS

2022– Assistant Professor, School of Government, Universidad Adolfo Ibáñez

2022– Young Researcher, Millennium Institute Foundational Research on Data (IMFD)

2026– Member, Editorial Committee, *Repositorio de Algoritmos Públicos*, GobLab UAI — national registry of automated decision systems in the Chilean public sector

2026– Member, Academic Core (Núcleo Académico), MSc in Economics and Public Policy (MEPP), UAI

2026– Member, Admissions Committee, PhD in Institutions and Political Processes, UAI (2027 intake)

EDUCATION

2014–20 PhD in Sociology, University of Cambridge. Thesis: "The Structure of Political Competition. Family Networks in the Chilean Congress, 1828–1894"

2017 Visiting Student Research Collaborator, Princeton University

2013–14 MPhil in Modern Society and Global Transformations, University of Cambridge — best-scored thesis of the cohort

2005–10 BA in Sociology, University of Chile — ranked third in cohort

1993–2002 Alliance Française, Curicó — national university entrance examination: best school score in mathematics and in language; best regional (Maule) score in history

FUNDED RESEARCH

As Principal Investigator

2025–28 **Fondecyt de Iniciación N° 11251464 (ANID).** "The Urban Elite Divide: A Geospatial Examination of the Brahmins vs. Merchants Hypothesis in Santiago, Chile." CLP 27,900,000. Sole PI; external, peer-reviewed.

2026–27 **PAI Interdisciplinary Associative Research, UAI.** "Open innovation methodologies with artificial intelligence applied to public challenges." CLP 8,000,000. PI, leading a team across three faculties (Government, Engineering and Sciences, Business). Awarded to 2 of 11 proposals; institutional evaluation 4.76/5.

2026–27 **Fondo Desafíos UAI (Engagement).** "Co-Lab Peñalolén" — AI decision-support tools co-designed with a municipal government. CLP 4,800,000. Director.

2024 **PAI Individual Research, UAI.** "Brahmins versus Merchants in Santiago, Chile? A geospatial approach." CLP 2,000,000. PI.

2021 **Development Bank of Latin America (CAF), research grant.** USD 8,000. PI.

As Co-Investigator

2024–28 **Fondecyt Regular N° 1241772 (ANID), PI Hans Lobel.** "Embed the City: An Artificial Intelligence-based Approach to Learn Transferable Representations from Multi-source and Multi-modal Urban Data." USD 60,000.

2022–26 **Fondecyt Regular N° 1220385 (ANID), PI Rodrigo Asún.** "Self-reinforcing mechanisms of protest cycles and social outbursts." USD 60,000.

Ongoing **Fondecyt Regular (ANID), PI Carmen Le Foulon.** "Checks & Balances at the Subnational Level: The Case of Chile."

Ongoing Projects (pilot or seeking funding)

2026– **AI Politicization Monitor** (with A. Abeliuk and M. Bosley). Weekly interviews of a citizen panel by an AI interviewer via WhatsApp, contrasted with LLM-simulated public opinion, to track how AI positions align with the left–right and establishment–anti-establishment axes. Pilot since October 2026.

2026– **Randomized controlled trial with Fondo Esperanza** (with A. Abeliuk). Effect of a WhatsApp AI assistant on the financial performance and business management of microentrepreneurs. Seeking international funding.

PEER-REVIEWED PUBLICATIONS

2026 **Bro N**, Luna JP. "Elitism in the Aisle: A Long-Run Surname Measure of Legislative Elite Composition in Chile, 1834–2020." *Research & Politics*, forthcoming (accepted October 2026). Preprint: arXiv:2607.08520. Replication data and code: github.com/naimbro/elitism-in-the-aisle

2026 Abeliuk A, Gaete V, **Bro N**. "Auditing socio-demographic and cross-societal fairness in LLM-simulated public opinion." *EPJ Data Science* (Q1). doi:10.1140/epjds/s13688-026-00673-y. Code and data: github.com/aabeliuk/fair-llm-surveys

2025 **Bro N**. "A frustratingly easy way of extracting political networks from text." *PLOS ONE*. doi:10.1371/journal.pone.0313149. Replication data published.

2024 **Bro N**, Morav L. "The economics of ethnic marriages: Endogamy and the social status of minority groups." *The British Journal of Sociology* 75(5), 830–853. doi:10.1111/1468-4446.13133

2024 Angles R, **Bro N**, Donoso-Guzmán I, Luna JP, Hogan A, Reutter J, Rosales-Méndez H, Toro S. "Telar and TelarKG: Data-Driven Insights into Chile's Constitutional Process." *Communications of the ACM*. doi:10.1145/3653312

2024 Reyes A, Mendoza M, Vera C, Lucchini F, Dimter J, Gutiérrez F, **Bro N**, Lobel H, Reyes A. "SpatialCluster: A Python library for urban clustering." *SoftwareX*. doi:10.1016/j.softx.2024.101739

2023 **Bro N**. "The Structure of Political Conflict. The Oligarchs and the Bourgeoisie in the Chilean Congress, 1834–1894." *Theory and Society*. doi:10.1007/s11186-022-09491-3. Top trending paper in sociology, ooir.org, July–August 2022.

2022 Vera C, Lucchini F, **Bro N**, Mendoza M, Lobel H, Gutiérrez F, Dimter J, Cuchacovic G, Reyes A, Valdivieso H, Alvarado N, Toro S. "Learning to cluster urban areas: two competitive approaches and an empirical validation." *EPJ Data Science*. doi:10.1140/epjds/s13688-022-00374-2

2022 Luna JP, Pérez C, Toro S, Rosenblatt F, Poblete B, Valenzuela S, Cruz A, **Bro N**, Alcatruz D, Escobar A. "Much Ado About Facebook? Evidence from 80 Congressional Campaigns in Chile." *Journal of Information Technology & Politics*. doi:10.1080/19331681.2021.1936334

2021 **Bro N**, Mendoza M. "Surname affinity in Santiago, Chile: A network-based approach that uncovers urban segregation." *PLOS ONE* 16(1): e0244372. doi:10.1371/journal.pone.0244372

2021 Mendoza M, **Bro N**. "Predicting affinity ties in a surname network." *PLOS ONE* 16(9): e0256603. doi:10.1371/journal.pone.0256603

WORKING PAPERS AND PREPRINTS

2026 **Bro N**, Abeliuk A, Fábrega J. "Graph-Based Ideal Points: Estimating Political Ideology from Signed Networks." Extended abstract accepted for oral presentation, Complex Networks 2026, Granada.

2026 Millán-Díaz H, **Bro N**. "Measuring Bureaucratic Reputation from Parliamentary Discourse: Chile's Electoral Service, 1926–2025." Under review, *Administration & Society*.

2021 Jaramillo-Echeverri J, Álvarez A, **Bro N**. "Surnames and Social Rank: Long-term Traits of Social Mobility in Colombia and Chile." CAF Working Papers.

SOFTWARE AND PUBLIC DATA

- **SpatialCluster** — open-source Python library for urban clustering (SoftwareX, 2024).
- **Simulated public-opinion forecasts** — weekly LLM-agent forecasting pipeline for Chile, published openly (2026–).
- **Replication dataset** for "A frustratingly easy way of extracting political networks from text" (PLOS ONE, 2025) — sufficient to reproduce both experiments in the paper.
- **Network and period datasets** for "The Structure of Political Conflict" (Theory & Society, 2023).

GRADUATE SUPERVISION

Doctoral — PhD in Institutions and Political Processes, UAI

- **Martín Bórquez** — principal supervisor (in progress).
- **Javiera Campos** — co-supervisor (in progress).
- **Héctor Millán** — co-supervisor (in progress).
- **Fabián Belmar** — thesis committee (2025).

Master's

- **Vanessa Gaete**, MSc in Data Science, University of Chile — external supervisor. "LLMs for Political Predictions in the Chilean Context" (completed 2024).

- **Matías Miranda**, MSc in Data Science, University of Chile — external supervisor. "Representations of Political Ideology in Language Models: A View of the Chilean Congress" (in progress).
- **Benjamín Palacios**, MSc in Data Science, UAI — co-supervisor. "Methods for identifying political cleavages in a news corpus" (in progress).
- **Maximiliano Osorio**, MSc in Economics and Public Policy, UAI — principal supervisor (in progress).

TEACHING

Graduate and undergraduate courses; computational and AI courses listed first. Syllabi at https://naimbro.github.io.

2026–27 **Machine Learning II** — MEPP, School of Government, UAI

2023–26 **AI in Research** — PhD in Institutions and Political Processes, UAI (2023, 2024, 2025, 2026); Programa ConectIA, UAI (2025); Faculty of Government, University of Chile (2025)

2026 **AI and Democracy** — Minor in Artificial Intelligence, UAI; intensive version, MEPP, UAI

2025–26 **Data Description and Visualization** — School of Government / MEPP, UAI

2025 **Computational Text Analysis for the Social Sciences** — Universidad Católica del Uruguay

2025 **AI and Public Policy** — Master in Public Policy, Universidad Católica del Uruguay

2023–25 **Social Media and Network Science** — School of Government and Faculty of Engineering and Sciences, UAI

2018–23 **Social Network Analysis** — MEPP, UAI (2022, 2023); Social Science Research Methods Centre, University of Cambridge (2018, 2019)

2024 **Topics in Legal Analytics** — Law School, UAI

2024 **Historical Social Networks** — Summer School in Mixed Methods, Pontificia Universidad Católica de Chile

2021–23 **Data and Public Policy** — UAI (2022, 2023); Diploma in Data Science for Public Policy, Pontificia Universidad Católica de Chile (2021, 2022)

2022 **Data Mining and Network Science** — MSc in Artificial Intelligence, Pontificia Universidad Católica de Chile

2020–22 **Big Data for the Social Sciences** — Department of Sociology, Universidad Diego Portales

2020–21 **Introduction to Computational Social Science** — Department of Sociology, University of Chile

2021–22 **Quantitative Sociology** — BIHE

2023–26 **Society, Culture and Politics** — Business School, UAI (2024, 2025, 2026); Quantitative Reasoning (2023); Elites in Contemporary Society, Faculty of Liberal Arts (2023)

2009–19 **Earlier teaching** — Supervisor, "Revolution, War, and Militarism," Department of Sociology, University of Cambridge (2014–2018); Teaching Assistant, Sociological Theory and Sociology of Development, University of Chile (2009–2012)

SELECTED CONFERENCE PRESENTATIONS

2027 **Congreso Futuro**, Chile — invited speaker (forthcoming)

2026 **Complex Networks 2026**, Granada, December — "Graph-Based Ideal Points: Estimating Political Ideology from Signed Networks" (with A. Abeliuk and J. Fábrega), oral presentation (forthcoming)

2026 **CENIA, National Center for Artificial Intelligence**, Human-Centric AI line (RL5), 6 July — invited talk, "Networks, Text and Ideology: Human-Centric AI"

2026 **ECPR Joint Sessions of Workshops**, Innsbruck, 7–14 April — "LLM-Extracted Political Networks as a Basis for Ideological Mapping: Evidence from Chilean Elites, 2019–2025"

2026 **SPSA Annual Meeting**, New Orleans, 14–17 January — "Digital Twins in Congress: Using Large Language Models to Predict Legislative Votes in Chile" (with A. Abeliuk, M. Bosley, J. P. Luna and S. Toro), panel "AI, LLMs, and Big Data: Applications"

2025 **Congreso Uruguayo de Ciencia Política**, Montevideo, 3–5 December — invited panel discussant

2024 **Complex Networks 2024**, Istanbul, 10–12 December — "Comparing LLMs for Extracting Political Networks from Text," oral presentation; poster "Surname Affinity Predicts Political Preference of Campaign Contributors in Chile" (with F. Zautzik and A. Hogan), selected for the conference's PLOS ONE special issue

2024 **IC2S2 2024**, Philadelphia, 17–20 July — "Extracting Political Networks from Text Using GPT-4" (with A. Abeliuk), poster

2023 **Complex Networks 2023**, Menton — "Sentiment-Aware Network Extraction from News Corpus using LLMs: An Empirical Validation with Legislative Agreement," oral presentation

2023 **SASE Annual Meeting**, Rio de Janeiro, 20–22 July — "Brahmins versus Merchants in Santiago, Chile? A Geospatial Approach," mini-conference "Elites and Power Structures"

2023 **IPSA World Congress of Political Science**, Buenos Aires, 15–19 July — "The Social Foundations of Political Discourse: Automatic Text Analysis of Chilean Legislative Speeches in a Time of Substantive Democratization, 1910–1950," panel "Elites and Democracy in Developing Nations"

2023 **Workshop on Corporate Power**, Universidad Diego Portales; Chilean Association for Social Network Analysis

2022 **American Sociological Association Annual Meeting**, Los Angeles

2021 Complex Networks; Networks 2021; Political Networks Conference; Urban Complex Systems; REPAL Annual Meeting, Cornell University; Jornadas Chilenas de Computación

2019–20 Nuffield Historical Mobility Conference, University of Oxford; Complex Networks 2020 (selected among 245 papers for a PLOS ONE special issue); SSHA Annual Meeting, Chicago; Sunbelt, Montreal; LASA, Boston

2014–18 LASA, Barcelona (travel grant); SSHA Annual Conferences, Montreal, Chicago (panel organiser), Baltimore and Toronto; COES Annual Conference, Santiago

AWARDS AND FELLOWSHIPS

2019 Berlin Summer School in Computational Social Sciences, travel grant, EUR 2,000

2018 Latin American Studies Association conference travel grant, USD 500

2017 St Catharine's College Thriplow Fund, GBP 5,000; Political Economy Society Supplementary Funding Competition, GBP 2,000

2016 Field research grant, Department of Sociology, University of Cambridge, GBP 6,000; Simón Bolívar Awards, Centre for Latin American Studies, GBP 1,661; Santander Research Grants, GBP 500

2013–17 CONICYT-Chile full doctoral fellowship, GBP 52,000

2014–16 St Catharine's College and Cambridge Sociology conference grants

2010–11 University of Chile research grants, PI (USD 1,500 each)

PUBLIC ENGAGEMENT AND POLICY

- **National media coverage of the LLM fairness research (2026):** TVN 24 Horas, El Mostrador, The Clinic, Radio Cooperativa, La Tercera (Radar), TrendTIC.
- **Opinion columns** in La Tercera and TerceraDosis on elites, AI and political change (2021–2026); earlier pieces in CIPER Chile, openDemocracy and Cambridge Globalist.
- **Public-sector collaboration** — AI decision-support tools co-designed with the Municipality of Peñalolén (2026–27).

LANGUAGES AND PROGRAMMING

Spanish (native); English, Persian and Portuguese (advanced); French (intermediate).

Programming: Python and R (advanced).